



```
// Master Table: AnonymizationRequests
{
  "PartitionKey": "ANON-2025-09",
  "RowKey": "REQ-{GUID}",
  "RequestId": "GUID",
  "RequestType": "Single|Bulk",
  "SubmittedBy": "user@domain.com",
  "SubmittedAt": "2025-09-09T10:00:00Z",
  "ScheduledFor": "2025-09-09T12:00:00Z",
  "TotalRecords": 5,
  "ProcessedRecords": 3,
  "SuccessfulRecords": 2,
  "FailedRecords": 1,
  "Status": "Pending|Processing|Completed|Failed",
  "FilePath": "blob://anonymization/requests/file.csv"
}
```

```
// Details Table: AnonymizationSteps
{
  "PartitionKey": "REQ-{GUID}",
  "RowKey": "{Timestamp}-{Phase}-{Step}",
  "CustomerId": "46469483",
  "AccountNumber": "6046463894591323",
  "Phase": "SQL|Cosmos|Databricks",
  "StepName": "UpdateCustomerTable",
  "Status": "Started|Completed|Failed",
  "StartTime": "2025-09-09T12:00:00Z",
  "EndTime": "2025-09-09T12:05:00Z",
  "ErrorDetails": "null",
  "RetryCount": 0
}
```

- High-level Databricks plan (per anonymize request)
- Trigger: Orchestrator calls Databricks Job (notebook) with param customerGuid (and requestId, mode=dryrun|apply, runId) after SQL & Cosmos steps completed for that customer.
 - Discovery: Notebook scans the tables
 - Backup: For each table to touch, create a backup snapshot or write a backup partition to a dr/recovery/ path. (Check with DE if needed)
 - Transform: For each table, produce anonymized rows (mask PII or write tombstone metadata). Use MERGE to update atomically.
 - Idempotency: Add/maintain metadata columns (anonymized_at, anonymized_by, anonymization_request_id, original_hash) and only re-anonymize if content hash/ChangeDatetime indicates change.
 - CDC Handling: Write tombstone markers (a _tombstone/deleted/anonymized flag column) or update CDC metadata so that streaming pipelines don't re-ingest original data. Confirm with DE team to ensure CDC jobs account for a deleted/ anonymized flag.
 - Audit: Append per-table per-customer audit record into an audit delta table anonymize_audit (with requestId, table, rowsAffected, start/end, status, jobRunId). Also upload detailed log to Blob.
 - Finish: Return job status to orchestrator; orchestrator updates master record.